Quantification: I534_z11c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	23
interior lacuna count	20
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 20)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2803.2	2018.5	1949.35	px^2
major_axis_length_px	lacuna shape	86.4528	83.0643	30.493	px
minor_axis_length_px	lacuna shape	47.4572	46.9029	20.5569	px
aspect_ratio	lacuna shape	1.935	1.9388	0.5632	unitless
eccentricity	lacuna shape	0.8085	0.8557	0.1165	unitless
solidity	lacuna shape	0.7198	0.7231	0.1205	unitless
roots_count	headline	11.5	10	5.3262	count
ring_length_r30_px	headline	309.75	299.5	118.538	px
ring_length_r60_px	reported	619.55	599	268.512	px
owned_length_px	ownership-dependent	754.394	625.742	575.378	px
edge_count	ownership-dependent	37.75	26	27.28	count
mean_edge_length_px	ownership-dependent	19.4949	19.7328	3.2599	px
perimeter_px	normalised	295.452	289.998	124.742	px
ring_area_r30_px2	normalised	9809.2	9932.5	3144.77	px^2
ring_area_r60_px2	normalised	20679.2	19972.5	6444.37	px^2
in_frame_fraction_r30	normalised	0.9837	1	0.0362	unitless
in_frame_fraction_r60	normalised	0.9618	1	0.0787	unitless
ring_density_r30	normalised	0.0326616	0.0320623	0.0101419	px^-1
ring_density_r60	normalised	0.0306183	0.0327602	0.0101982	px^-1
roots_per_100px_perimeter	normalised	4.0256	4.1609	0.9629	per 100 px
ring_attached_length_r30_px	network v2	286.7	270.5	121.731	px
ring_attached_length_r60_px	network v2	490.65	487	268.236	px
ring_length_w_r30_px	network v2	352.937	341.235	137.326	px
ring_length_w_r60_px	network v2	703.781	669.473	308.29	px
sholl_crossings_r10	network v2	7.1	6	2.8266	count
sholl_crossings_r20	network v2	7.75	7	3.7539	count
sholl_crossings_r30	network v2	8.55	8.5	3.9666	count
loop_count	loops, report only	2.05	1	3.0689	count
loop_edge_share	loops, report only	0.1155	0.0594	0.1335	share
ring_width_mean_r30_px	width	7.699	7.721	0.7426	px

Per field
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measure	value	units
analysed_area_px2	787122	px^2
total_skeleton_length_px	22868	px
canalicular_length_density_per_px	0.0290527	px^-1
junction_count	498	count
junction_density_per_px2	0.00063268	px^-2
lacuna_count	23	count

lacunae_per_px2	2.922e-05	px^-2
skeleton_component_count	273	count
mean_component_length_px	83.7656	px
median_component_length_px	24	px
field_length_density_w_per_px	0.0330145	px^-1
field_density_without_flagged_per_px	0.0233697	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	148	px
unattached_component_count	10	count
canalicular_length_density_attached_per_px	0.0290527	px^-1
field_length_density_with_unattached_per_px	0.0292407	px^-1
excluded_area_px2	193353	px^2
analysed_field_area_px2	855223	px^2
loop_count	53	count
loop_edge_share	0.1363	share
edge_count	1365	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.1674	px
width_median_px	8.2462	px
width_p10_px	4	px
width_p90_px	11.6619	px
width_pixels_used	23416	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	5018	113.612	66.0072	1.7212	0.8139	0.7134	-1.5671	33.3908	243.731	True
2	1429	75.9455	34.6321	2.1929	0.89	0.5788	0.9821	47.7124	162.336	False
3	1748	63.7004	52.4126	1.2154	0.5683	0.625	1.3536	36.7677	399.808	False
4	2001	74.781	36.9486	2.0239	0.8694	0.794	-1.1924	101.66	153.378	False
5	4592	90.8676	67.3599	1.349	0.6712	0.8423	-1.2953	137.617	765.059	False
6	902	70.1443	23.2471	3.0173	0.9435	0.6677	-1.2472	150.552	535.413	False
7	2333	67.9441	45.1639	1.5044	0.7471	0.9554	-0.4874	234.38	1000.91	True
8	697	50.8222	20.9969	2.4205	0.9107	0.7383	1.293	287.073	212.017	False
9	3532	87.4182	55.1855	1.5841	0.7756	0.7638	1.235	329.842	344.026	False
10	4129	107.816	50.0459	2.1543	0.8857	0.9077	-1.34	378.661	498.241	False
11	889	60.4502	27.413	2.2052	0.8913	0.6247	-1.2388	363.547	49.8526	False
12	1735	68.9793	37.2098	1.8538	0.842	0.6893	-0.903	463.884	594.474	False
13	751	42.1527	25.0435	1.6832	0.8044	0.8632	-1.1817	459.581	482.401	False
14	5417	114.374	70.6956	1.6178	0.7861	0.6168	1.316	527.301	55.7192	True
15	3037	99.839	58.2931	1.7127	0.8118	0.571	1.4007	558.019	301.395	False
16	4073	115.2	49.8465	2.3111	0.9015	0.7391	0.3504	603.966	867.145	False
17	4016	127.91	62.2247	2.0556	0.8737	0.6106	1.1858	619.072	672.485	False
18	7591	130.597	95.8386	1.3627	0.6793	0.6789	-0.2363	683.477	82.1523	False
19	4140	142.999	43.9592	3.253	0.9516	0.7236	-1.1238	670.561	384.423	False
20	554	30.7716	24.9383	1.2339	0.5858	0.895	1.0673	729.574	697.603	False
21	1972	94.2757	63.7697	1.4784	0.7365	0.4735	0.8655	817.668	378.753	False
22	2036	78.7104	35.4441	2.2207	0.8929	0.8879	-1.002	828.682	94.3777	False

23 6240 115.676 84.3355 1.3716 0.6844 0.7226 -1.0938 881.946 274.251 False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	15	337	443	568.174	29	19.5922	6.7223
2	11	264	350	288.108	21	13.7194	6.4414
3	9	352	597	847.37	50	16.9474	7.3539
4	10	277	601	177.853	13	13.681	6.6277
5	15	443	812	1321.09	68	19.4278	7.575
6	8	322	834	1427.45	72	19.8258	7.7452
7	6	155	358	401.421	19	21.1274	8.7355
8	7	199	419	426.262	23	18.5331	8.5102
9	16	500	1055	2184.78	102	21.4194	8.5149
10	11	341	760	1172.19	60	19.5365	8.3787
11	8	192	330	232.836	10	23.2836	6.0936
12	12	360	671	745.453	29	25.7053	7.7108
13	5	155	258	175.125	10	17.5125	8.2917
14	19	351	508	723.027	34	21.2655	5.9354
15	19	385	862	1396.18	70	19.9455	7.8658
16	14	408	1045	1085.1	51	21.2766	7.1052
17	23	503	1001	1288.79	58	22.2204	7.4999
18	23	499	869	903.436	46	19.6399	7.5211
19	9	207	349	267.61	17	15.7418	7.6147
20	5	175	403	152.882	7	21.8403	7.7313
21	7	154	273	165.711	11	15.0646	9.1306
22	8	232	483	506.031	22	23.0014	8.1991
23	10	227	419	323.622	15	21.5748	8.0692

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	494.351	8231	12179	0.7421	0.6708	0.0409428	0.0363741	3.0343
2	282.291	4862	7142	0.9455	0.8134	0.0542986	0.0490059	3.8967
3	270.635	8169	16049	0.8695	0.7737	0.0430897	0.0371986	3.3255
4	236.936	6970	15047	1	1	0.0397417	0.0399415	4.2205
5	341.505	11471	28225	1	1	0.0386191	0.0287688	4.3923
6	176.167	7690	20930	1	1	0.0418726	0.0398471	4.5412
7	199.782	5730	14247	0.6804	0.6335	0.0270506	0.0251281	3.0033
8	136.125	6663	16736	1	1	0.0298664	0.0250358	5.1423
9	297.706	11030	23297	1	1	0.0453309	0.0452848	5.3744
10	305.605	10599	19545	1	1	0.0321729	0.0388846	3.5994
11	191.167	7524	16666	0.9189	0.7962	0.0255183	0.0198008	4.1848
12	238.451	9266	19550	1	1	0.0388517	0.0343222	5.0325
13	122.953	5564	11136	1	1	0.0278577	0.0231681	4.0666
14	490.399	11727	20563	0.8061	0.7376	0.0299309	0.0247046	3.8744
15	378.475	12733	27257	1	1	0.0302364	0.0316249	5.0201
16	373.546	12922	30830	1	1	0.0315741	0.0338956	3.7479
17	515.428	13936	28482	1	1	0.0360936	0.035145	4.4623
18	555.955	15261	25089	0.9401	0.8614	0.0326977	0.0346367	4.137
19	383.889	12892	25468	1	1	0.0160565	0.0137035	2.3444
20	97.9828	5477	13592	1	1	0.0319518	0.0296498	5.1029
21	335.12	10722	19872	1	1	0.014363	0.0137379	2.0888
22	222.693	8786	20073	1	0.9917	0.0264056	0.0240622	3.5924
23	446.416	13647	28597	1	1	0.0166337	0.0146519	2.2401

Network v2 per lacuna

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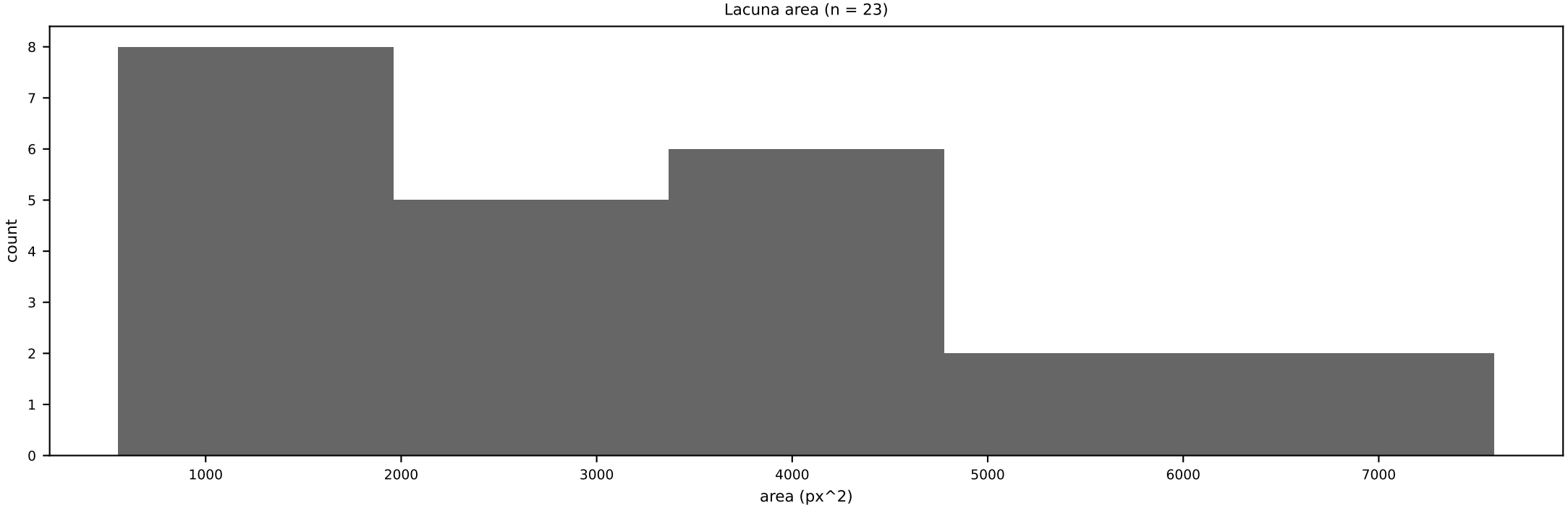
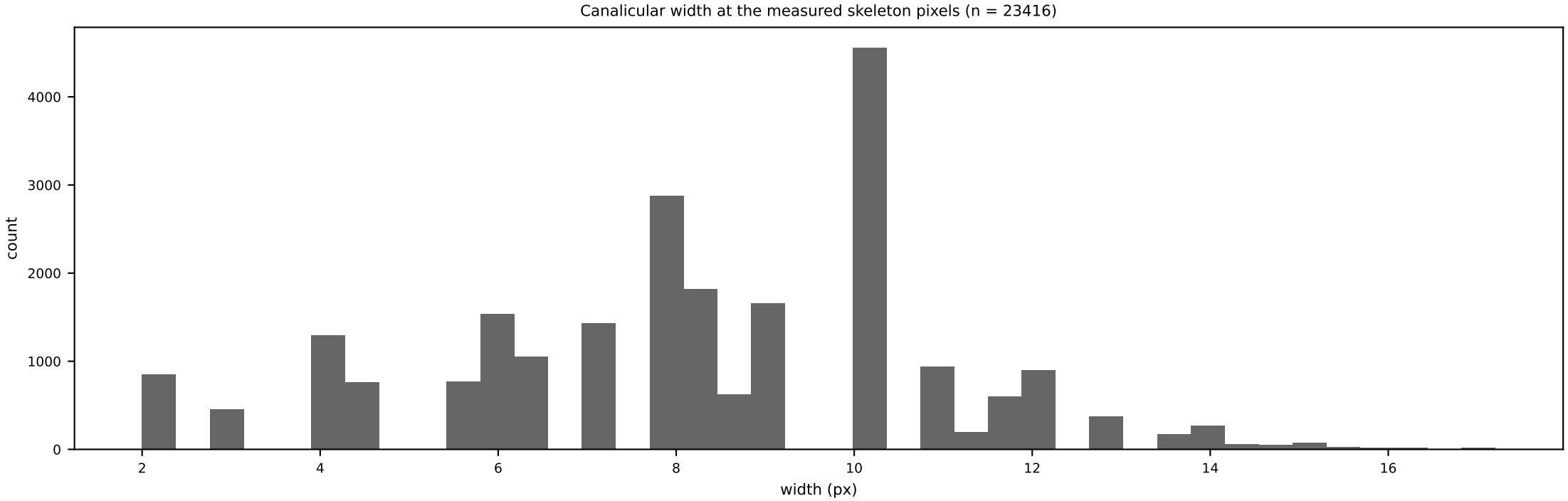
lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	334	421	381.806	497.647	9	8	4
2	257	292	296.279	394.706	9	5	3
3	348	545	392.836	660.688	6	9	6
4	179	179	314.907	678.257	6	4	8
5	423	720	507.274	922.051	11	11	13
6	284	643	367.563	955.779	6	11	10
7	140	252	165.669	393.836	3	4	4
8	192	283	230.137	479.375	5	6	3
9	486	1025	568.688	1201.18	10	16	18
10	327	691	395.505	874.566	8	9	12
11	148	172	222.936	387.103	5	2	5
12	340	596	412.605	779.867	6	7	9
13	145	178	175.297	287.551	4	4	4
14	327	352	403.576	592.541	7	3	5
15	376	722	438.233	975.264	11	10	11
16	395	767	467.961	1172.69	8	11	12
17	492	842	577.014	1147.92	10	10	11
18	451	748	578.399	1005.9	13	14	13
19	199	281	226.853	381.007	5	4	7
20	150	279	193.711	457.304	4	6	5
21	129	169	173.61	301.936	2	3	5
22	232	429	267.794	549.902	6	6	7
23	181	252	251.137	462.576	7	7	9

I534_z11c1-2: distributions, pre-validation, px